

B55H02**High Density Polyethylene****Blow Molding****Product Description:**

HDPE B55H02 is a natural colored grade produced with the latest Mitsui CX slurry polymerization Technology exhibiting following features:

- ✓ High Impact Strength
- ✓ Rigidity
- ✓ Low Parison Sag
- ✓ Good ESCR Properties

HDPE B55H02 is recommended for following applications:

- ✓ Large volume containers for transport & storage of Chemicals, Detergent, Lube Oils etc.

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 5.00 kg)	ASTM D1238	g / 10 min	0.20
2	Melt Flow Index (190°C & 21.6 kg)	ASTM D1238	g / 10 min	5.00
3	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.955
Mechanical Properties				
4	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	28
5	Tensile Strength @ Break	ASTM D638	MPa	>27
6	Elongation @ Break (50mm / min)	ASTM D638	%	>600
7	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	500
8	Charpy Impact Strength	ISO 179-1	KJ/m ²	NB
9	Flexural Modulus	ASTM D790A	MPa	1000
10	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>600
11	Hardness	ASTM D2240	Shore D	64
Thermal Properties				
12	Vicat Softening Point (10N)	ASTM D1525	°C	125
13	Heat Deflection Temperature (0.455 Mpa)	ASTM D648	°C	80
14	Melting Temperature	ASTM D3418	°C	132
** Mechanical Properties tested on Compression Molded Specimen. * Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Provisional Technical Datasheet

B55H02



Regulatory Requirements:

HDPE Blow Molding grade B55H02 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



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B55H03**High Density Polyethylene****Blow Molding****Product Description:**

HDPE B55H03 is a natural colored grade produced with the latest Mitsui CX slurry polymerization Technology exhibiting following features:

- ✓ High Impact Strength
- ✓ Rigidity
- ✓ Low Parison Sag
- ✓ Good ESCR Properties

HDPE B55H03 is recommended for following applications:

- ✓ Large Containers (up to 100L) for transport & storage of Chemicals, Detergent, Lube Oil etc.
- ✓ Blow Molded Water Tanks (up to 1000L)

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 5.00 kg)	ASTM D1238	g / 10 min	0.35
2	Melt Flow Index (190°C & 21.6 kg)	ASTM D1238	g / 10 min	9.00
3	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.955
Mechanical Properties				
4	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	28
5	Tensile Strength @ Break	ASTM D638	MPa	40
6	Elongation @ Break (50mm / min)	ASTM D638	%	>600
7	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	500
8	Charpy Impact Strength	ISO 179-1	KJ/m ²	NB
9	Flexural Modulus	ASTM D790A	MPa	1000
10	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>600
11	Hardness	ASTM D2240	Shore D	64
Thermal Properties				
12	Vicat Softening Point (10N)	ASTM D1525	°C	129
13	Heat Deflection Temperature (0.455 Mpa)	ASTM D648	°C	80
** Mechanical Properties tested on Compression Molded Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Provisional Technical Datasheet

B55H03



Regulatory Requirements:

HDPE Blow Molding grade B55H03 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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B5004**High Density Polyethylene****Blow Molding****Product Description:**

HDPE B5004 is a natural colored grade produced with the latest Mitsui CX slurry polymerization

Technology exhibiting following features:

- ✓ Good Processing Properties
- ✓ Excellent ESCR Properties
- ✓ Good external appearance

HDPE B5004 is recommended for following applications:

- ✓ Bottles & containers up to 5 L
- ✓ Packaging of Pesticides & Surfactants
- ✓ Bottles for Shampoo and Cosmetics

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	0.40
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.950
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	24
4	Tensile Strength @ Break	ASTM D638	MPa	>15
5	Elongation @ Break (50mm / min)	ASTM D638	%	>500
6	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	140
7	Charpy Impact Strength	ISO 179-1	KJ/m ²	8.5
8	Flexural Modulus	ASTM D790A	MPa	900
9	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>500
10	Hardness	ASTM D2240	Shore D	65
Thermal Properties				
11	Vicat Softening Point (10N)	ASTM D1525	°C	121
12	Heat Deflection Temperature (0.455 Mpa)	ASTM D648	°C	78
** Mechanical Properties tested on Compression Molded Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 160 – 200 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under

PWM (Plastics Waste Management) Rules, 2016.

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B5004



Regulatory Requirements:

HDPE Blow Molding grade B5004 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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B6003**High Density Polyethylene****Blow Molding****Product Description:**

HDPE B6003 is a natural colored grade produced with the latest Mitsui CX slurry polymerization

Technology exhibiting following features:

- ✓ Very Good Processing Properties
- ✓ High Rigidity
- ✓ Excellent Mechanical properties

HDPE B6003 is recommended for following applications:

- ✓ General purpose containers up to 20 L
- ✓ Lubricant oil containers (up to 15 L)
- ✓ Edible Oil containers (up to 15 L)
- ✓ Bottles for food, milk packaging

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	0.35
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.962
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	32
4	Elongation @ Break (50mm / min)	ASTM D638	%	>500
5	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	350
6	Flexural Modulus	ASTM D790A	MPa	1300
7	Hardness	ASTM D2240	Shore D	68
Thermal Properties				
8	Vicat Softening Point	ASTM D1525	°C	129
9	Heat Deflection Temperature	ASTM D648	°C	80
** Mechanical Properties tested on Compression Molded Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 160 – 200 °C

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B6003



Regulatory Requirements:

HDPE Blow Molding grade B6003 shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

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F52H04



High Density Polyethylene

Film Extrusion

Product Description:

HDPE F52H04 is a natural colored grade produced with the latest Mitsui CX slurry polymerization Technology exhibiting following features:

- ✓ Excellent Mechanical Strength
- ✓ High Stiffness
- ✓ Excellent processability (at High Speed)

HDPE F52H04 is recommended for following applications:

- ✓ Tissue-like films
- ✓ Garment/Grocery/merchandise bags
- ✓ Disposal waste bags
- ✓ Shopping/Counter bags

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 5.00 kg)	ASTM D1238	g / 10 min	0.25
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.952
Mechanical Properties				
3	Tensile Strength @ Yield	ASTM D638	MPa	24
4	Elongation @ Break	ASTM D638	%	>500
5	Notched Izod Impact Strength	ASTM D256	J/m	315
6	Charpy Impact Strength	ISO 179-1	KJ/m ²	NB
7	Hardness	ASTM D2240	Shore D	64
8	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>500
Thermal Properties				
9	Vicat Softening Point (10N)	ASTM D1525	°C	124
** Mechanical Properties tested on Compression Molded type IV Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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F52H04



Regulatory Requirements:

HDPE Film grade F52H04 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

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PARC/2022/06 - 04

M5220



High Density Polyethylene

Injection Molding

Product Description:

HDPE M5220 is a natural colored grade produced with the latest Ineos Gas Phase polymerization

Technology exhibiting following features:

- ✓ Good Flow
- ✓ High Warpage Resistance
- ✓ Suitable for faster molding cycles

HDPE M5220 is recommended for following applications:

- ✓ Housewares, storage bins
- ✓ Thin wall food containers
- ✓ Caps & Closures
- ✓ Base resin for masterbatch

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	21
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.952
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	22.0
4	Elongation @ Break (50mm / min)	ASTM D638	%	350
5	Flexural Modulus	ASTM D790	MPa	900
6	Izod Impact Strength (at 23 °C)	ASTM D256	J/m	30
7	Hardness	ASTM D2240	Shore D	62
Thermal Properties				
8	Vicat Softening Point (10N)	ASTM D1525	°C	125
9	Heat Deflection Temperature (0.455 MPa)	ASTM D648	°C	80
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 190 – 230 °C

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M5220



Regulatory Requirements:

HDPE grade M5220 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

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Health and Safety Information:

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M6008



High Density Polyethylene

Injection Molding

Product Description:

HDPE M6008 is a natural colored grade produced with the latest Ineos Gas Phase polymerization

Technology exhibiting following features:

- ✓ Excellent Mechanical Properties
- ✓ Very Good processability
- ✓ Good dimensional stability
- ✓ Good aesthetics of finished products

HDPE M6008 is recommended for following applications:

- ✓ Material handling crates
- ✓ Helmets
- ✓ Caps & Closures
- ✓ Luggage shells

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	8.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.960
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	28.0
4	Elongation @ Break (50mm / min)	ASTM D638	%	>500
5	Flexural Modulus	ASTM D790	MPa	850
6	Flexural Yield Strength	ASTM D790	MPa	20
7	Izod Impact Strength (at 23 °C)	ASTM D256	J/m	80
8	Charpy Impact Strength	ISO 179-1	KJ/m ²	4.0
9	Hardness	ASTM D2240	Shore D	65
Thermal Properties				
10	Vicat Softening Point (10N)	ASTM D1525	°C	128
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 190 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under

PWM (Plastics Waste Management) Rules, 2016.

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M6008



Regulatory Requirements:

HDPE grade M6008 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



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M6008U**High Density Polyethylene****Injection Molding****Product Description:**

HDPE M6008U is a natural colored grade produced with the latest Ineos Gas Phase polymerization

Technology exhibiting following features:

- ✓ Excellent Mechanical Properties
- ✓ Good dimensional stability
- ✓ Good aesthetics of finished products
- ✓ UV Stabilized

HDPE M6008U is recommended for following applications:

- ✓ Material handling crates, Pellets
- ✓ Pails
- ✓ Helmets
- ✓ Luggage shells

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	8.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.960
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	28.0
4	Elongation @ Break (50mm / min)	ASTM D638	%	>700
5	Flexural Modulus	ASTM D790	MPa	900
6	Flexural Yield Strength	ASTM D790	MPa	20
7	Izod Impact Strength (at 23 °C)	ASTM D256	J/m	85
8	Charpy Impact Strength	ISO 179-1	KJ/m ²	6.0
9	Hardness	ASTM D2240	Shore D	65
Thermal Properties				
10	Vicat Softening Point (10N)	ASTM D1525	°C	128
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 190 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Provisional Technical Datasheet

M6008U



Regulatory Requirements:

HDPE grade M6008U meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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High Density Polyethylene

Pipe Extrusion

Product Description:

HDPE P50H03 is a natural colored grade produced with the latest Mitsui CX slurry polymerization Technology exhibiting following features:

- ✓ Very Good Processing Properties
- ✓ High Creep Resistance
- ✓ Excellent Stress Crack Resistance
- ✓ Excellent mechanical properties

HDPE P50H03 is recommended for following applications:

- ✓ High pressure Pipe applications (PE-100)
- ✓ Potable water transportation pipes.
- ✓ Higher Diameter pipes for drainage/ Sewerage applications.

MRS Classification: OPaLene HDPE P50H03 has a Minimum required strength (MRS) classification of 10 MPa according to ISO 9080 and is designated PE100 grade according to ISO 12162.

Properties	Test Method	UOM	Minimum Burst time
SCG (Notch pipe testing @ 80°C, 9.2 bar)	ISO 13479	Hrs	>500
RCP- S4 test (Critical pressure to Rapid crack propagation)	ISO 13477	Bar	≥10 (Critical Pressure @ 0°C)

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 5.0 kg)	ASTM D1238	g / 10 min	0.27
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.950
Mechanical Properties				
3	Tensile Strength @ Yield	ASTM D638	MPa	23
4	Elongation @ Yield	ASTM D638	%	10
5	Elongation @ Break	ASTM D638	%	>800
6	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	>250
7	Charpy Impact Strength	ISO 179-1	KJ/m ²	NB
8	Flexural Modulus	ASTM D790A	MPa	880
9	Hardness	ASTM D2240	Shore D	60
10	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>1000
Thermal Properties				
11	Vicat Softening Point (10N)	ASTM D1525	°C	125
12	Heat Deflection Temperature (0.455 Mpa)	ASTM D648	°C	65
13	Oxidative Induction Time (OIT)	ASTM D3895	Minutes	>30
** Mechanical Properties tested on Compression Molded type IV Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under

PWM (Plastics Waste Management) Rules, 2016.

P50H03



Regulatory Requirements:

HDPE Pipe grade P50H03 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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P51H10



High Density Polyethylene

Pipe Extrusion

Product Description:

HDPE P51H10 is a natural colored grade produced with the latest Mitsui CX slurry polymerization Technology exhibiting following features:

- ✓ Very Good Processing Properties
- ✓ Good mechanical properties
- ✓ Excellent Stress Crack Resistance
- ✓ Good Creep Resistance

HDPE P51H10 is recommended for following applications:

- ✓ Pressure pipes for drinking water and Irrigation
- ✓ Pipes for Civil Engineering & Drainage
- ✓ Corrugated Tubes & Pipes.
- ✓ Large Diameter irrigation pipes

MRS Classification: OPaLene HDPE P51H10 has a Minimum required strength (MRS) classification of 10 MPa according to ISO 9080 and is designated PE100 grade according to ISO 12162.

Properties	Test Method	UOM	Minimum Burst time
SCG (Notch pipe testing @ 80°C, 8.0 bar)	ISO 13479	Hrs	430
RCP- S4 test (Critical pressure to Rapid crack propagation)	ISO 13477	Bar	≥10 (Critical Pressure@ 0°C)

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 5 kg)	ASTM D1238	g / 10 min	0.45
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.951
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	23
4	Elongation @ Break (50mm / min)	ASTM D638	%	>700
5	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	>200
6	Flexural Modulus	ASTM D790A	MPa	850
7	Hardness	ASTM D2240	Shore D	60
8	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>600
Thermal Properties				
9	Vicat Softening Point (10N)	ASTM D1525	°C	122
10	Heat Deflection Temperature (0.455 Mpa)	ASTM D648	°C	62
11	Oxidative Induction Time (OIT)	ASTM D3895	Minutes	>30
** Mechanical Properties tested on Compression Molded Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under

PWM (Plastics Waste Management) Rules, 2016.

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P51H10



Regulatory Requirements:

HDPE Pipe grade P51H10 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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PARC/2022/06 - 04

P5002



High Density Polyethylene

Pipe Extrusion

Product Description:

HDPE P5002 is a natural colored grade produced with the latest Mitsui CX slurry polymerization

Technology exhibiting following features:

- ✓ Very Good Processing Properties
- ✓ Good mechanical properties
- ✓ Excellent Stress Crack Resistance
- ✓ Good external appearance

HDPE P5002 is recommended for following applications:

- ✓ PE-63 pressure pipes for drinking water & Irrigation
- ✓ Steel pipe lining
- ✓ Ducting

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 5.00 kg)	ASTM D1238	g / 10 min	0.90
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.950
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	20
4	Elongation @ Break (50mm / min)	ASTM D638	%	>500
5	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	>200
6	Flexural Modulus	ASTM D790A	MPa	900
7	Hardness	ASTM D2240	Shore D	60
8	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D1693B	Hours	>600
Thermal Properties				
9	Vicat Softening Point (10N)	ASTM D1525	°C	120
10	Heat Deflection Temperature (0.455 Mpa)	ASTM D648	°C	65
11	Oxidative Induction Time (OIT)	ASTM D3895	Minutes	>30
** Mechanical Properties tested on Compression Molded Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Provisional Technical Datasheet

P5002



Regulatory Requirements:

HDPE Pipe grade P5002 shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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R5410



High Density Polyethylene

Raffia & Monofilament

Product Description:

HDPE R5410 is a natural colored grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Exceptional Tenacity
- ✓ Excellent processability

HDPE R5410 is recommended for following applications:

- ✓ Woven Sacks & Tarpaulin
- ✓ Knitted nets for crop protection
- ✓ “ Raschel ” sacks for vegetables
- ✓ Ropes

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	0.90
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.954
Mechanical Properties				
3	Tensile Strength @ Yield	ASTM D638	MPa	25
4	Tensile Strength @ Break	ASTM D638	MPa	35
5	Elongation @ Break	ASTM D638	%	>500
6	Charpy Impact Strength	ISO 179-1	KJ/m ²	12
7	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	250
8	Flexural Modulus	ASTM D790A	MPa	850
9	Hardness	ASTM D2240	Shore D	63
Thermal Properties				
10	Vicat Softening Point (10N)	ASTM D1525	°C	122
** Mechanical Properties tested on Compression Molded type IV Specimen. * Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 200 – 240 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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R5410



Regulatory Requirements:

HDPE raffia grade R5410 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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Y5710



High Density Polyethylene

Monofilament

Product Description:

HDPE Y5710 is a natural colored grade produced with the latest Ineos Gas Phase polymerization

Technology exhibiting following features:

- ✓ Exceptional Tenacity
- ✓ Excellent processability

HDPE Y5710 is recommended for following applications:

- ✓ Monofilament for Nets
- ✓ Twines, Ropes

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g / 10 min	0.90
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.957
Mechanical Properties				
3	Tensile Strength @ Yield	ASTM D638	MPa	24
4	Tensile Strength @ Break	ASTM D638	MPa	35
5	Elongation @ Break	ASTM D638	%	>500
6	Charpy Impact Strength	ISO 179-1	KJ/m ²	8
7	Notched Izod Impact Strength (at 23 °C)	ASTM D256A	J/m	120
8	Flexural Modulus	ASTM D790A	MPa	950
9	Hardness	ASTM D2240	Shore D	63
Thermal Properties				
10	Vicat Softening Point (10N)	ASTM D1525	°C	122
** Mechanical Properties tested on Compression Molded type IV Specimen.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 200 – 240 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Y5710



Regulatory Requirements:

HDPE monofilament grade Y5710 meets the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade meets the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



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E2507



Linear Low Density Polyethylene

Extrusion Coating

Product Description:

LLDPE E2507 is a butene co-monomer based linear low density grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Processability
- ✓ Good adhesion properties.

- ✓ Excellent Tear strength & abrasion resistance.

LLDPE E2507 is recommended for following applications:

- ✓ Extrusion Coating on HDPE Woven Fabrics, Aluminum foil, Jute, Paper etc.

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	7.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.927
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	10
4	Elongation @ Break (50mm / min)	ASTM D638	%	450
5	Flexural Modulus	ASTM D 790A	MPa	200
6	Notched Izod Impact Strength	ASTM D 256A	J/m	510
7	Co-efficient of Friction on films (Static/Dynamic)	ASTM D 1894	-	0.80/0.70
8	Vicat Softening Point (10N)	ASTM D 1525	°C	85
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 220 – 250 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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E2507



Regulatory Requirements:

LLDPE grade E2507 shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



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PARC/2022/06 - 03

F2001A

Linear Low Density Polyethylene

Packaging Films

Product Description:

LLDPE F2001A is a butene co-monomer based natural colored grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Mechanical Properties
- ✓ Very Good optical properties
- ✓ Easy Processing
- ✓ Without Slip & Anti-block additives

LLDPE F2001A is recommended for following applications:

- ✓ Co-extrusion films
- ✓ Lamination films
- ✓ Drip laterals
- ✓ Agriculture Films

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	0.90
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.918
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min) (MD/TD)	ASTM D882	MPa	11.0/12.0
4	Tensile Strength @ Break (50mm / min) (MD/TD)	ASTM D882	MPa	44.0/40.0
5	Elongation @ Break (50mm / min) (MD/TD)	ASTM D882	%	600/800
6	Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709	g/μm	3.5
7	Tear Strength (MD/TD)	ASTM D1922	g/μm	3.3/10.3
** Mechanical Properties tested on 40 μ (microns) thick monolayer blown film made with 1.8 mm die gap and 2.25 BUR.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 170 – 210 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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F2001A



Regulatory Requirements:

LLDPE film grade F2001A shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent PE Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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F2001S**Linear Low Density Polyethylene****Packaging Films****Product Description:**

LLDPE F2001S is a butene co-monomer based natural colored grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Mechanical Properties
- ✓ Very Good optical properties
- ✓ Easy Processing
- ✓ With Slip & Anti-block additives

LLDPE F2001S is recommended for following applications:

- ✓ Co-extrusion films
- ✓ Liquid Packaging films
- ✓ Drip laterals
- ✓ Agriculture Films

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	0.90
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.919
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min) (MD/TD)	ASTM D882	MPa	11.0/12.0
4	Tensile Strength @ Break (50mm / min) (MD/TD)	ASTM D882	MPa	44.0/40.0
5	Elongation @ Break (50mm / min) (MD/TD)	ASTM D882	%	600/800
6	Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709	g/μm	3.5
7	Tear Strength (MD/TD)	ASTM D1922	g/μm	3.5/10.0
8	Coefficient of Friction (Static/Dynamic)	ASTM D1894	-	0.23/0.16
9	Film Gloss (60°)	ASTM D523	%	70
** Mechanical Properties tested on 40 μ (microns) thick monolayer blown film made with 1.8 mm die gap and 2.25 BUR.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 170 – 210 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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F2001S



Regulatory Requirements:

LLDPE film grade F2001S shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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F2002A

Linear Low Density Polyethylene

Packaging Films

Product Description:

LLDPE F2002A is a butene co-monomer based natural colored grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Good Mechanical Properties
- ✓ Low gels
- ✓ Easy Processing

- ✓ No Slip & Anti-block additives

LLDPE F2002A is recommended for following applications:

- ✓ Lamination films
- ✓ Stretch cling films
- ✓ Air bubble films
- ✓ Drip Laterals

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	2.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.920
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min) (MD/TD)	ASTM D882	MPa	11.0/12.0
4	Tensile Strength @ Break (50mm / min) (MD/TD)	ASTM D882	MPa	35.0/30.0
5	Elongation @ Break (50mm / min) (MD/TD)	ASTM D882	%	800/750
6	Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709	g/μm	3.0
7	Tear Strength (MD/TD)	ASTM D1922	g/μm	3.25/7.5
** Mechanical Properties tested on 40 μ (microns) thick monolayer blown film made with 1.8 mm die gap and 2.25 BUR.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 170 – 210 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Provisional Technical Datasheet

F2002A



Regulatory Requirements:

LLDPE film grade F2002A shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

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F2002S**Linear Low Density Polyethylene****Packaging Films****Product Description:**

LLDPE F2002S is a butene co-monomer based natural colored grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Good Mechanical Properties
- ✓ Very Good optical properties
- ✓ Easy Processing

- ✓ With Slip & Anti-block additives

LLDPE F2002S is recommended for following applications:

- ✓ General purpose packaging films
- ✓ Blending with LDPE/LLDPE

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	2.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.920
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min) (MD/TD)	ASTM D882	MPa	11.0/12.0
4	Tensile Strength @ Break (50mm / min) (MD/TD)	ASTM D882	MPa	35.0/30.0
5	Elongation @ Break (50mm / min) (MD/TD)	ASTM D882	%	800/750
6	Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709	g/μm	3.0
7	Tear Strength (MD/TD)	ASTM D1922	g/μm	3.25/7.5
8	Coefficient of Friction (Static/Dynamic)	ASTM D1894	-	0.23/0.16
9	Film Gloss (60°)	ASTM D523	%	70
** Mechanical Properties tested on 40 μ (microns) thick monolayer blown film made with 1.8 mm die gap and 2.25 BUR.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 170 – 210 °C

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F2002S



Regulatory Requirements:

LLDPE film grade F2002S shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

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Linear Low Density Polyethylene

Injection Molding

Product Description:

LLDPE M2525 is a butene co-monomer based linear low density grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Flow & Processability
- ✓ Good warpage resistance.

LLDPE M2525 is recommended for following applications:

- ✓ Housewares
- ✓ Thin walled containers & Lids
- ✓ Suitable for faster molding cycle.
- ✓ Masterbatch & Compounding

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	25.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.934
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	12
4	Elongation @ Break (50mm / min)	ASTM D638	%	>500
5	Flexural Modulus	ASTM D 790A	MPa	250
6	Vicat Softening Point (10N)	ASTM D 1525	°C	95
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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M2525



Regulatory Requirements:

LLDPE grade M2525 shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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M2550



Linear Low Density Polyethylene

Injection Molding

Product Description:

LLDPE M2550 is a butene co-monomer based linear low density grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Flow & Processability
- ✓ Good warpage resistance.

LLDPE M2550 is recommended for following applications:

- ✓ Masterbatch & Compounding
- ✓ Housewares
- ✓ Thin walled containers & Lids
- ✓ Suitable for faster molding cycle.

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	50.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.934
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	13
4	Elongation @ Break (50mm / min)	ASTM D638	%	>400
5	Flexural Modulus	ASTM D 790A	MPa	400
6	Vicat Softening Point (10N)	ASTM D 1525	°C	95
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 220 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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M2550



Regulatory Requirements:

LLDPE grade M2550 shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



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T3804



Linear Low Density Polyethylene

Rotational Moulding

Product Description:

LLDPE T3804 is a butene co-monomer based medium density grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Processability
- ✓ High Stiffness & Impact Strength
- ✓ Excellent Internal & External Surface finish

LLDPE T3804 is recommended for following applications:

- ✓ Overhead water tanks
- ✓ Loft tanks
- ✓ Storage bins
- ✓ Toys and traffic barriers
- ✓ Sanitation & Agriculture products

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	4.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.938
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	15
4	Elongation @ Break (50mm / min)	ASTM D638	%	>500
5	Flexural Modulus	ASTM D 790A	MPa	280
6	Notched Izod Impact Strength	ASTM D 256A	J/m	650
7	ESCR (F ₅₀ , 10% Igepal Soln. v/v)	ASTM D 1693B	Hours	200
8	Vicat Softening Point (10N)	ASTM D 1525	°C	117
* Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 230 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under

PWM (Plastics Waste Management) Rules, 2016.

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T3804



Regulatory Requirements:

LLDPE grade T3804 shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



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T3804U



Linear Low Density Polyethylene

Rotational Moulding

Product Description:

LLDPE T3804U is a butene co-monomer based medium density grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Processability
- ✓ UV (HALS) Stabilized
- ✓ High Stiffness & Impact Strength
- ✓ Excellent Internal & External Surface finish

LLDPE T3804U is recommended for following applications:

- ✓ Overhead water tanks
- ✓ Storage bins
- ✓ Toys and traffic barriers
- ✓ Sanitation & Agriculture products
- ✓ Rotomolding applications requiring outdoor exposure

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (190°C & 2.16 kg)	ASTM D1238	g/10 min	4.0
2	Density (at 23 °C)	ASTM D1505	gm/cm ³	0.938
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D638	MPa	15
4	Elongation @ Break (50mm / min)	ASTM D638	%	>500
5	Flexural Modulus	ASTM D 790A	MPa	280
6	Notched Izod Impact Strength	ASTM D 256A	J/m	650
7	ESCR (F ₅₀ ,10% Igepal Soln. v/v)	ASTM D 1693B	Hours	200
8	Vicat Softening Point (10N)	ASTM D 1525	°C	117
** Mechanical Properties tested on Injection molded specimen in accordance with ASTM D4101 and conditioned as per ASTM D618.				
* Typical Values and not to be taken as specification limits, values may change without any prior notice.				

Recommended Processing Temperature: 180 – 230 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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T3804U



Regulatory Requirements:

LLDPE grade T3804U shall meet the requirements stipulated in IS 10146:1982 on 'Specification of Polyethylene for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10141:1982. The Grade and the additives incorporated in it shall also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent Polyethylene Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

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FH10S

Polypropylene Homo Polymer

Tubular Water Quench Films

Product Description:

PP Homo polymer FH10S is a natural colored high stiffness grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Higher flow for easy processing.
- ✓ High Stiffness.
- ✓ Excellent gloss & clarity of films
- ✓ With Slip & Antiblock additives

PP Homo Polymer FH10S is recommended for following applications:

- ✓ Textile Packaging
- ✓ Snack Food Packaging
- ✓ Grocery Packaging
- ✓ General Purpose Packaging
- ✓ Bread Bags

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (230°C & 2.16 kg)	ASTM D1238	g/10 min	10
2	Density (23 °C)	ASTM D 1505	gm/cm ³	0.90
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D 638	MPa	35
4	Elongation @ Break (50mm / min)	ASTM D 638	%	30
5	Flexural Modulus (1.3 mm/min)	ASTM D790A	MPa	1500
6	Notch Izod Impact Strength (23°C)	ASTM D 256	J/m	25
7	Hardness (Rockwell)	ASTM D 785	R Scale	95
Thermal Properties				
8	Heat Deflection Temperature (0.46N/m ²)	ASTM D648	°C	100
** Mechanical Properties tested on Injection molded specimen prepared in accordance with ASTM D 4101 and conditioned as per ASTM D 618				
* Typical Values and not to be taken as specifications, values may change without any prior notice.				

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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Provisional Technical Datasheet

FH10S



Regulatory Requirements:

PP Homo Polymer grade FH10S shall meet the requirements stipulated in IS 10910 on 'Specification for Polypropylene and its Copolymers for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10909. The Grade and the additives incorporated in it will also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent PP Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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Polypropylene Homo Polymer

Injection Molding

Product Description:

PP Homo polymer MH13 is a natural colored high stiffness grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Higher flow for easy processing.
- ✓ High Stiffness.
- ✓ Good dimensional stability
- ✓ Faster Molding cycle

PP Homo Polymer MH13 is recommended for following applications:

- ✓ Household articles
- ✓ Consumer Products
- ✓ Rigid Packaging

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (230°C & 2.16 kg)	ASTM D1238	g/10 min	13.5
2	Density (23 °C)	ASTM D 1505	gm/cm ³	0.90
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D 638	MPa	34
4	Elongation @ Yield (50mm / min)	ASTM D 638	%	6
5	Flexural Modulus (1.3 mm/min)	ASTM D790A	MPa	1150
6	Notch Izod Impact Strength (23°C)	ASTM D 256	J/m	20
7	Hardness (Rockwell)	ASTM D 785	R Scale	95
Thermal Properties				
8	Vicat Softening Point (10N)	ASTM D1525	°C	150
9	Heat Deflection Temperature (0.46N/m ²)	ASTM D648	°C	100
** Mechanical Properties tested on Injection molded specimen prepared in accordance with ASTM D 4101 and conditioned as per ASTM D 618.				
* Typical Values and not to be taken as specifications, values may change without any prior notice.				

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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MH13



Regulatory Requirements:

PP Homo Polymer grade MH13 shall meet the requirements stipulated in IS 10910 on 'Specification for Polypropylene and its Copolymers for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10909. The Grade and the additives incorporated in it will also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent PP Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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MH13A**Polypropylene Homo Polymer****Injection Molding****Product Description:**

PP Homo polymer MH13A is a natural colored high stiffness grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Higher flow for easy processing.
- ✓ High Stiffness.
- ✓ Good dimensional stability
- ✓ Anti-Static Property

PP Homo Polymer MH13A is recommended for following applications:

- ✓ Household articles
- ✓ Furniture Products
- ✓ Rigid Packaging
- ✓ Compounding

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (230°C & 2.16 kg)	ASTM D1238	g/10 min	12
2	Density (23 °C)	ASTM D 1505	gm/cm ³	0.90
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D 638	MPa	35
4	Elongation @ Break (50mm / min)	ASTM D 638	%	200
5	Flexural Modulus (1.3 mm/min)	ASTM D790A	MPa	1400
6	Notch Izod Impact Strength (23°C)	ASTM D 256	J/m	25
7	Hardness (Rockwell)	ASTM D 785	R Scale	95
Thermal Properties				
8	Vicat Softening Point (10N)	ASTM D1525	°C	150
9	Heat Deflection Temperature (0.46N/m ²)	ASTM D648	°C	100
** Mechanical Properties tested on Injection molded specimen prepared in accordance with ASTM D 4101 and conditioned as per ASTM D 618. * Typical Values and not to be taken as specifications, values may change without any prior notice.				

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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MH13A



Regulatory Requirements:

PP Homo Polymer grade MH13A shall meet the requirements stipulated in IS 10910 on 'Specification for Polypropylene and its Copolymers for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10909. The Grade and the additives incorporated in it will also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent PP Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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Polypropylene Homo Polymer

Woven Sacks

Product Description:

PP Homo Polymer RH03 is a natural colored high performance grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Very good processability.
- ✓ Good balance of tenacity & elongation.
- ✓ Low water carry-over.

PP Homo Polymer RH03 is recommended for following oriented structures (Stretched Tapes) applications:

- ✓ Cement Packaging bags
- ✓ Fertilizer Packaging bags
- ✓ Food Grain Packaging bags
- ✓ Bulk Industrial Packaging (FIBC)
- ✓ General purpose wrapping fabric

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (230°C & 2.16 kg)	ASTM D1238	g/10 min	3.3
2	Density (23 °C)	ASTM D 1505	gm/cm ³	0.90
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D 638	MPa	35
4	Elongation @ Yield (50mm / min)	ASTM D 638	%	6
5	Flexural Modulus (1.3 mm/min)	ASTM D790A	MPa	1350
6	Notch Izod Impact Strength (23°C)	ASTM D 256	J/m	44
7	Hardness (Rockwell)	ASTM D 785	R Scale	100
Thermal Properties				
8	Vicat Softening Point (10N)	ASTM D1525	°C	150
9	Heat Deflection Temperature (0.46N/m ²)	ASTM D 648	°C	95
** Mechanical Properties tested on Injection molded specimen prepared in accordance with ASTM D 4101 conditioned as per ASTM D 618. * Typical Values and not to be taken as specifications, values may change without any prior notice.				

Recommended Processing Temperature: 200 – 260 °C

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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RH03



Regulatory Requirements:

PP Homo Polymer grade RH03 shall meet the requirements stipulated in IS 10910 on 'Specification for Polypropylene and its Copolymers for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10909. The Grade and the additives incorporated in it will also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent PP Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

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RH25X

Polypropylene Homo Polymer

Fibers & Filaments

Product Description:

PP Homo polymer RH25X is a natural colored high performance grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ Excellent Processability & strength
- ✓ Very good resin purity & uniformity
- ✓ Anti-gas fading properties

PP Homo Polymer RH25X is recommended for following applications:

- ✓ Multi filament yarns
- ✓ Fine denier staple fibers
- ✓ Spun bond non-woven

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (230°C & 2.16 kg)	ASTM D1238	g/10 min	25
2	Density (23 °C)	ASTM D 1505	gm/cm ³	0.90
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D 638	MPa	35
4	Elongation @ Break (50mm / min)	ASTM D 638	%	20
5	Flexural Modulus (1.3 mm/min)	ASTM D790A	MPa	1450
6	Notch Izod Impact Strength (23°C)	ASTM D 256	J/m	20
7	Hardness (Rockwell)	ASTM D 785	R Scale	95
Thermal Properties				
8	Heat Deflection Temperature (0.46N/m ²)	ASTM D648	°C	100
** Mechanical Properties tested on Injection molded specimen prepared in accordance with ASTM D 4101 and conditioned as per ASTM D 618				
* Typical Values and not to be taken as specifications, values may change without any prior notice.				

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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RH25X



Regulatory Requirements:

PP Homo Polymer grade RH25X shall meet the requirements stipulated in IS 10910 on 'Specification for Polypropylene and its Copolymers for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10909. The Grade and the additives incorporated in it will also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

Storage & Handling:

Prevent PP Material from direct exposure to sunlight & heat to avoid quality deterioration. The storage location should be dry, dust free and the Storage temperature should not exceed 50 °C. Non - compliance to these precautionary measures can lead to degradation of the product causing Color changes, Odor & inadequate product performance.

Health and Safety Information:

The product described herein may require precautions in handling and use because of toxicity, flammability, or other consideration. The Material Safety Data Sheet (MSDS) contains the available product health and safety information for this material and can be found at www.opalindia.in. Before using any material, a customer is advised to consult the MSDS for the product under consideration for use.



ONGC Petro additions Ltd

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Contact: ONGC Petro Additions Ltd., 35, Nutan Bharat Co-operative Housing Society Limited,, R.C. Dutt Road, Alkapuri, Vadodara-390007, Gujarat India, Telephone: +91 265 6192600, Fax: +91 265 6192622, Corporate Site: www.opalindia.in PARC/2022/06 - 01

Polypropylene Homo Polymer

Non-Woven & Fibers

Product Description:

PP Homo polymer RH38 is a natural colored high performance grade produced with the latest Ineos Gas Phase polymerization Technology exhibiting following features:

- ✓ High Melt flow
- ✓ Excellent Processability
- ✓ Very good resin purity & uniformity

- ✓ Anti-gas fading properties

PP Homo Polymer RH38 is recommended for following applications:

- ✓ Fine denier fibers
- ✓ Spun bond non-woven
- ✓ High speed spinning

Typical Properties:

Sr. No.	Properties	Test Method	Units	Values*
Physical Properties				
1	Melt Flow Index (230°C & 2.16 kg)	ASTM D1238	g/10 min	38
2	Density (23 °C)	ASTM D 1505	gm/cm ³	0.90
Mechanical Properties				
3	Tensile Strength @ Yield (50mm / min)	ASTM D 638	MPa	31
4	Elongation @ Yield (50mm / min)	ASTM D 638	%	6.5
5	Flexural Modulus (1.3 mm/min)	ASTM D790A	MPa	1085
6	Notch Izod Impact Strength (23°C)	ASTM D 256	J/m	20
7	Hardness (Rockwell)	ASTM D 785	R Scale	95
Thermal Properties				
8	Heat Deflection Temperature (0.46N/m ²)	ASTM D648	°C	100
** Mechanical Properties tested on Injection molded specimen prepared in accordance with ASTM D 4101 conditioned as per ASTM D 618. * Typical Values and not to be taken as specifications, values may change without any prior notice.				

Note: This product is not to be used in the manufacture of SUP (Single Use Plastic) items prohibited under PWM (Plastics Waste Management) Rules, 2016.

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RH38



Regulatory Requirements:

PP Homo Polymer grade RH38 shall meet the requirements stipulated in IS 10910 on 'Specification for Polypropylene and its Copolymers for safe use in contact with Foodstuff, Pharmaceutical & Drinking water'. The grade and Additives incorporated in this grade shall meet the positive list of constituents as prescribed in IS 10909. The Grade and the additives incorporated in it will also comply with the FDA: CFR Title 21,177.1520, Olefin Polymers.

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